

Tuesday Jan 14 (Environment Setup Cont'd)

1) Verify Unix environment is set up

Mac OS: Default environment should be ok. Test the Unix-based commands ('ls', 'cd', ...) and C compilation/execution to make sure they're working (covered later)

Windows: We're using **WSL** (Windows Subsystem for Linux) with Ubuntu distribution (default). From your Windows terminal (Start + cmd or Powershell, running as Admin)

- wsl --install
- Alternate: wsl --install -d Ubuntu
- Alternate: Install Ubuntu from Microsoft Store

After successful install, execute the following from Windows terminal (cmd/Powershell), confirm that Ubuntu is present

- wsl -l -v
- wsl --status
- wsl --list --online

Entering WSL through terminal:

- wsl
 - Opens your current Windows directory, so you can use Unix-based commands
- ubuntu
 - Opens WSL home directory (could be more useful for assignments)

Leave WSL:

- exit

2) VS Code Setup

Mac OS:

- Go to 'Extensions' on the side bar and install C/C++
- Unix and C commands should already work in your terminal (next part)

Windows:

- Go to 'Extensions' on the side bar and look for and install WSL (from Microsoft)
- Click on blue box in the bottom left and connect to remote host: WSL
 - Now you are "inside" your Linux subsystem; click on 'Explorer' on side bar, and press OK to navigate your Linux files from your home directory
 - Install C/C++ extension in WSL
 - Make sure your Linux subsystem is updated and install compiler (gcc) for C. **apt** is our package manager for Ubuntu
 - sudo apt update && sudo apt upgrade

- `sudo apt install build-essential`
 - `gcc --version`
 - `(sudo apt install gdb)`
- To disconnect from WSL, click on the blue box and select 'Close Remote Connection' from the dropdown to return to Windows file system

3) Basic Workflow in VS Code and Terminal

- Try compiling + executing a sample C file (like `hello.c` from the internet) either using the terminal or through VS Code interface
 - VS Code: Select "Debug C/C++ file" and/or "Run C/C++ file"
 - Terminal:
 - `gcc hello.c -o hello` → compiles `hello.c` into executable file called 'hello'
 - `./hello` → executes 'hello'

Thursday Jan 16 (Intro to Terminal/Unix Commands)

Refer to `commandLines.pdf`